

Azure Virtual Desktop & Application Delivery Architecture

Cloud Infrastructure Blueprint | Student ID: 2400031215 | K L University

1. Multi-Tier Architecture Blueprint

Architecture Tier	Deployed Component	Technical Role & Interconnect
Client Access	Windows App / HTML5 Client	Encrypted outbound TLS 1.2 over TCP 443 to Microsoft Global Network
AVD Control Plane (PaaS)	Host Pool: hp-avd-engineering-lab App Group: dag-avd-engineering-lab Workspace: ws-avd-engineering-lab	Brokers incoming user sessions, manages pooled load balancing (B...
Identity & RBAC	Microsoft Entra ID (kluniversity.in) Group: AVD-Engineering-Lab-Students	Authenticates student identities, grants Desktop Virtualization User p...
Virtual Network Layer	VNet: vnet-avd-lab (10.0.0.0/16) Subnet: snet-avd (10.0.0.0/24) NSG: nsg-avd-lab	Isolates session hosts with zero public IP exposure. Outbound rules...
Compute Layer	Session Host: vm-avd-sh-0 SKU: Standard_B4ms (4 vCPU, 16 GB) OS: Windows 11 Multi-Session 23H2	Hosts multi-user desktop sessions, runs engineering software, automa...
Profile Storage Layer	Storage: stavdprofiles260916 Share: fslogix (128 GiB, 1026 IOPS) Lock: lock-storage-prevent-delete	Decouples user profiles via dynamic VHDX containers mounted ove...

2. Application Delivery & Gateway Architecture

A critical architectural requirement is understanding how application delivery operates in Azure Virtual Desktop compared to traditional Web Application Gateways:

- **Traditional Web Apps (Reference Project P10):** Employs an *Azure Application Gateway with WAF* to terminate HTTP/HTTPS traffic at Layer 7 and inspect requests using OWASP core rule sets before forwarding to backend web pools.
- **Virtual Desktop Application Delivery (This Project):** Utilizes the *AVD Reverse Connect Gateway*. Session hosts initiate outbound connections to Azure PaaS gateways on port 443. Inbound traffic from students is authenticated at Microsoft Entra ID and brokered directly to the session host without exposing any inbound ports or requiring a public IP on the VM.
- **Dual Delivery Modes:** Supports both **Full Desktop** delivery (via Desktop Application Groups) and **RemoteApps** (seamless application streaming of individual engineering tools).

3. Security & Data Protection Controls

- **Zero Public Exposure:** No public IP address or inbound RDP port 3389 opened to the internet.
- **CanNotDelete Resource Lock:** Applied to the storage account to safeguard profile containers against administrative misclicks.
- **Automated Cost Guardrail:** DevTestLabs shutdown task enforces compute stoppage outside laboratory testing hours.